



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.C.A. (CBCS) Degree Examinations

May - June 2024

22MCA204 – DATA WAREHOUSING AND DATA MINING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full question** from each unit.

Unit – I

		Marks	BT*	CO*	PO*
1.	a) Briefly discuss about multitier architecture of Datawarehouse.	5	L*1	1	1
	b) Give details about Datawarehouse models.	5	L1	1	1
	c) Elaborate extraction, transformation and loading with specific architecture	10	L2	1	1
2.	a) With examples illustrate stars and snowflakes schema.	10	L2	1	1
	b) Discuss typical OLAP operations with examples.	10	L2	1	1

Unit – II

3.	a) Compute SMC and Jaccard coefficient for the following two binary vectors X= (1,0,0,0,0,0,0,0,0,0) Y= (0,0,0,0,0,0,1,0,0,1)	5	L2	2	2
	b) Differentiate between discrete and continuous attributes with examples.	5	L1	2	1
	c) Describe dimensionality reduction and sampling with an example.	10	L1	2	1
4.	a) Illustrate that data mining is an integral part of knowledge discovery in databases.	5	L2	2	2
	b) Data mining is a confluence of many disciplines, discuss.	5	L2	2	1
	c) Describe different types of ordered data with examples.	10	L2	2	1

Unit – III

5.	a) Discuss the Alternative methods for generating frequent itemsets.	10	L1	3	1
	b) Illustrate why use support and confidence in association analysis.	5	L2	3	2
	c) What is candidate generation and candidate pruning? Discuss.	5	L1	3	1
6.	a) Illustrate frequent itemset generation in the Apriori algorithm with suitable example.	10	L2	3	2
	b) Discuss the steps of F-P tree construction by considering the following transaction data set, min support = 2.				

TID	Items
1	{a,b}
2	{b,c,d}
3	{a,c,d,e}
4	{a,d,e}
5	{a,b,c}
6	{a,b,c,d}
7	{a}
8	{a,b,c}
9	{a,b,d}
10	{b,c,a}

10 L3 3 3

P.T.O

Unit – IV

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|----|--|----|----|---|---|
| 7. | a) What is confusion matrix? Discuss with example. | 5 | L2 | 4 | 1 |
| | b) Define accuracy and error rate with formula. | 5 | L2 | 4 | 2 |
| | c) Illustrate Hunts algorithm using training set for predicting borrowers who will default on loan payments. | 10 | L4 | 4 | 2 |
| 8. | a) Explain Nearest -Neighbor classifier with example. | 10 | L2 | 4 | 1 |
| | b) How Bayes theorem can be used for classification? Elaborate with example. | 10 | L2 | 4 | 2 |

Unit – V

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|-----|--|----|----|---|---|
| 9. | a) Define and discuss some of the squared error (SEE). | 5 | L1 | 5 | 1 |
| | b) Elaborate basic K-means algorithm. | 5 | L2 | 5 | 2 |
| | c) Illustrate classification-oriented measures of cluster validity. | 10 | L2 | 5 | 2 |
| 10. | a) Elaborate agglomerative hierarchical clustering algorithm with example. | 10 | L1 | 5 | 1 |
| | b) Illustrate DBSCAN is a simple and effective density-based clustering algorithm with the concept of core, border and noise points. | 10 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
